

Divyansh Yadav

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Department of Mechanical Engineering

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023 - Present	B.Tech	Indian Institute of Technology, Kanpur	6.8/10
2022	CBSE(XII)	Radiant Central Academy, Ambedkar Nagar	87.4%
2020	CBSE(X)	VSS Educational Academy, Ambedkar Nagar	87.8%

Scholastic Achievements

- Secured **All India Rank 5592** in **JEE Advanced 2023** among the 2.5 lakh shortlisted candidates
- Secured **All India Rank 15354** in **JEE Mains 2023** among the 1.1 million candidates

Key Projects

PDF ChatBot – RAG-based Document Q&A System (May'26)
Self Project | 🐙

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline enabling users to upload PDF documents and ask natural language questions with accurate, context-grounded responses in real time
- Implemented document parsing using **PyMuPDF**, recursive chunking, **sentence-transformer** embeddings with **ChromaDB** and a hybrid retriever combining **BM25** keyword search with semantic similarity matching
- Developed a **FastAPI** backend with **LangChain** orchestration and **Groq** hosted **Llama-3.3-70B**, along with a responsive **Streamlit** frontend for real-time chat and source attribution

Implemented GPT-2 Architecture from Scratch and Fine-tuned for Spam Classification (Jul'25)
Self Project | 🐙

- Built a transformer-based **GPT-2 language model** from scratch in **PyTorch**, including **multi-head attention**, **positional encoding** and **layer normalization**
- Trained the model on a custom dataset to validate implementation correctness
- Integrated OpenAI's pre-trained GPT-2 weights and fine-tuned the model for spam classification, achieving improved accuracy and performance compared to baseline methods
- Gained hands-on experience in **LLM architectures**, **transfer learning**, and **fine-tuning techniques**, applying them effectively to solve a range of downstream NLP tasks

AES-128 Implementation and Side-Channel Attack (CPA) | 🐙 (May'25-Jul'25)
Course Project | CS698G | Prof. Udaya Parampalli

- Implemented the **AES-128 encryption algorithm** from scratch in Python, including **key expansion**, **Sub-Bytes**, **ShiftRows**, **MixColumns** and **AddRoundKey**
- Designed and executed a **Correlation Power Analysis (CPA)** attack on the first encryption round, analyzing simulated power traces to successfully recover individual cryptographic key bytes
- Demonstrated that cryptographic systems can be compromised through side-channel vulnerabilities, even when the underlying algorithms are mathematically strong and theoretically secure
- Gained hands-on experience in **cryptanalysis**, **side-channel attacks** and **secure systems design**

Technical Skills

- Programming Languages:** Python, C, C++, Bash, $\text{\LaTeX}$
- Libraries & Frameworks:** PyTorch, LangChain, FastAPI, Streamlit, Scikit-learn, Pandas, NumPy, Matplotlib, Sentence-Transformers
- Tools & Platforms:** Git, GitHub, ChromaDB, Jupyter

Relevant Courses

Fundamentals of Data Science and Machine Intelligence Applied Cryptography Linear Algebra Single Variable Calculus Introduction to Electronics	Software Development and Operations Fundamentals of Computing Application of Single Variable and Multivariable Calculus Differential Equations Robotics
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